SAMPLE

Cosmos Unveiled: Cosmic Strings

Cracks in the Fabric of Space

Aira Astralis

Free sample: Introduction & Chapter 1
putrabooks.github.io

Introduction

Among the ideas the twentieth century's physics left for cosmology, few are as strange, or as instructive, as the cosmic string. The concept proposes that the universe, in its first moments, may have acquired cracks: one-dimensional faults in space itself, thinner than a proton, longer than galaxies, under tensions that no material could bear, and that some of these cracks may still exist, stretched across the sky, waiting to be found. The chapters tell that idea's story, from its origin in the particle physicists' equations, through its rise as a candidate for the universe's structure, to its fall from that role and its modern life as a testable prediction of fundamental physics.

The idea rewards a full telling because it sits at a junction of the century's deepest theories. Cosmic strings come from the same mathematics that the particle physicists built to unify the forces, from the study of what happens when a symmetry breaks, and they belong, at the other end, to the largest scales astronomy measures, the shaping of galaxies and the furthest light. A single cosmic string, if one exists, would be a fossil of the universe's opening instants, a hands-on specimen of physics at energies no accelerator will ever touch, and the search for them is therefore one of the few ways observation can reach the physics beyond the standard model of particles.

The reader should hold the scale's fewness, because it defines the subject's company. The physics beyond the standard model, the particle theorists' umbrella for everything the established theory does not contain, is tested in three ways in the modern era: the great accelerators, which push energies upward at colossal cost and diminishing reach, the rare processes, which hunt for forbidden decays and suppressed interactions in mountains of data, and the relics, which wait for the universe's own first second to have left evidence lying around, and the string searches belong to the third way, the cheapest per proton-equivalent and the most patient, running on telescope time and compute rather than on tunnel-diameter steel. The reader who follows the three ways' budgets will find the relic way's patience financed, in the string case, by instruments built for other prey, the searches riding the great machines' spare capacity, a bargain that keeps the wager affordable indefinitely. The relic way's virtue is its reach: a fossil from the grand-unified scale samples energies a trillion times the accelerators' frontier, and the virtue's price is the wager, that the young universe dropped something findable, and the reader now knows the strings' version of the wager whole.

The book's plan is straightforward. The opening chapters introduce the object: what a cosmic string is, how the early universe made it, and how a whole network of them should behave. The middle chapters are the physics: the way a string bends space, the images it would leave in the sky, and the gravitational waves its loops should shed. The later chapters are the history: the strings' rise as the favored theory of structure formation, the observations that retired them from that role, the superstring revival that gave the idea a second life, and the searches, in lensing, in the microwave sky, and in the new gravitational-wave detectors, that continue as this book is written.

A word on the book's posture, said here once and honored throughout: the pages ahead describe a live subject, not a settled one. No cosmic string has ever been observed, and the observational limits have grown tight enough to rule out the simplest versions of the idea as the main source of the universe's structure. But the limits themselves are the story's substance: the searches have real sensitivity, the theory makes real predictions, and the subject, unlike much of the physics beyond the standard model, is falsifiable in principle and daily in practice, and the reader will see, chapter by chapter, how such a subject behaves.

One reading-map note before the physics begins, on the book's chapters and their independence. The reader who wants the object's physics first can read the opening chapters and stop, with the strings' nature and behavior complete, and the reader who wants the drama can jump to the history chapters, the promise and the verdict, which stand alone as a case study, and the middle physics chapters, the lensing and the waves, are the observational heart, and the final chapters, the searches and the limits, are the book's live news. The book was built to be read whole, but its doors open separately, and the reader's own curiosity should choose the entrance, because the subject, like the strings themselves, has no preferred direction through it. The physics, at every entrance, is the same physics, and the reader's time is the only currency spent.

One framing image, before the physics begins: a cosmic string is to space what a crease is to cloth, a line where the fabric's smoothness failed, left behind when the whole cloth changed its weave at once. The universe's first moments, on the theories this book describes, were exactly such a change, and the creases, if the theories are right, are still there, and finding one would be like finding the seam of the sky.

The reader should also know, at the outset, where the subject sits in the larger project of this book's series. The cosmos unveiled by modern astronomy is mostly the study of what the universe contains, stars and galaxies and

the radiation between them, and the cosmic string belongs to a smaller, stranger shelf: the study of what the universe is, the structure of space and the fields that fill it, the questions that lie beneath the inventory. The string searches are therefore kin to the gravitational-wave detections and the particle-collider programs rather than to the galaxy surveys, and the book's emphasis follows the kinship: the reader will find less catalog and more logic here, more about what a detection would mean than about what has been seen, because the subject itself is a bet on the universe's deepest grammar, and the bet's terms, not the sky's census, are its substance.

One more framing note before the physics, on the book's treatment of numbers, because the subject's quantities deserve the reader's honest handling. The strings' tension, the fences' positions, and the energy scales are all stated in this book in powers of ten, rounded, because the book's purpose is the shape of the subject rather than its ledger's last digit, and the reader who wants the current exact limits will find them, precisely dated and cited, in the subject's reviews, and the reader should note the practice's honesty: a popular book that quoted the fences to their third digit would be lying by precision, since the limits move with every instrument run, and the book's rounding is the same courtesy the subject's own papers extend to the public, the significant figures kept honest by the moving data, and anyone who finishes the volume will hold the numbers to the accuracy the subject itself warrants, which is the only accuracy worth having.

Chapter 01: The Object: A Crack in Space

The reader should begin with the object itself, stated as plainly as the physics allows, because a cosmic string is unlike anything in everyday experience and the description must be built carefully, one property at a time.

A cosmic string is a one-dimensional object: it has length but no width in any meaningful sense. The strings the theory proposes are extraordinarily thin, with a thickness set by the energy scale of the physics that made them, and for the standard cosmic strings the scale is so far beyond experiment that a convenient comparison is to the proton: a string would be to a proton as a proton is to a human hair, roughly, a hierarchy that means a cosmic string is, for all astronomical purposes, a line with no thickness at all. The string's length, by contrast, can be cosmic: the theory predicts both closed loops, which can be as small as the theory permits or as large as a galaxy, and infinitely long strings, which would stretch across the visible universe.

The string's substance is not matter but energy, and the distinction matters. A cosmic string is a concentration of the energy of a broken symmetry, a kind of trapped field, and its mass, if one can use the word, is distributed entirely along its length, a fixed amount per unit length, which the physicists call the tension and write with the symbol mu. The tension of the strings the theory favors would be enormous, the mass of a mountain range per kilometer of string, and the reader should let the image settle: a thread, invisible by thickness, weighing like a mountain chain, coiling through the cosmos.

The string's strangest property is the one that makes it interesting: it does not end. The mathematics that describes the strings forbids them from simply terminating in space, so every string is either a closed loop or part of an infinite line, and strings interact only by intersecting themselves or one another, exchanging partners or pinching off loops when they cross. The prohibition is not a rule imposed on the strings but a consequence of what they are, the same way a soap bubble's surface cannot simply end in mid-air, and the reader should hold the analogy: the strings are like defects in a crystal, and a defect's geometry is dictated by the order it violates.

The chapter's closing property is gravitational, and it is the seed of everything the later chapters describe. A cosmic string bends space around itself in a way no ordinary object does: the space near a straight string is conical, like a sheet of paper with a wedge cut out and the edges pasted together, and the reader who walks a circle around the string finds the circle shorter than geometry says it should be. The wedge is the string's whole gravitational effect, and the deficit angle, in the theories of interest, would be tiny, seconds of arc, but the effect is pure geometry: the string carries no Newtonian pull toward itself, and an object at rest near a straight string feels no attraction, which is the property that sets the strings apart from every star and planet and makes their searches, as the reader will see, a distinctive kind of astronomy.

The reader should test the conical picture with a small thought experiment, because the test cements the physics. Two light rays, passing a straight string on opposite sides, arrive at the observer having traveled through the pasted-together space, and each ray's path is straight in its own geometry, but the wedge's removal makes the two paths converge as though bent by a lens, and the wedge's angle is the whole effect, with the string's thickness playing no part at all. The reader who wants the number's scale can hold a comparison: a string at the classic grand-unified tension would bend the two paths by a few seconds of arc, the apparent width of a coin seen across a city, a tiny angle that the whole sky's worth of string would nevertheless paint, and the smallness is the searches'

central difficulty and the theory's central prediction, all in one number.

The conical picture's laboratory test deserves its paragraph, because the reader should know the geometry has been checked on Earth. The defect mechanics the theory uses, the pinning and the crossing rules, has been tested in the condensed-matter analogues, the liquid crystals, the superfluids, and the superconductors, where the same mathematics produces visible defects, the vortex lines in a superfluid, the disclinations in a crystal's fluid phase, and the analogues' networks, simulated and observed in the laboratory, scale as the cosmological theory predicts, the same self-regulation at bench scale that the sky's network would show at cosmic scale. The reader should hold the test's meaning honestly: the laboratory confirms the mathematics of defect formation and network evolution, not the existence of cosmic strings, which remain a cosmological hypothesis, and the confirmation's value is the confidence it licenses, that when the theory says the early universe froze and cracked, the freezing and cracking it describes are physics, seen by eye in other media, and the cosmic case differs only in scale, and in the waiting for the sky's answer.

The kitchen analogy earns a careful upgrade, because the physics is stranger than the analogy admits. A crack in ice marks where the bonding failed; a cosmic string is not the absence of material but the presence of trapped energy, a tube of false vacuum a planck-scale hair wide and, per unit length, heavier than a mountain range, with no ends because it either closes on itself or spans the observable universe. The comparison to a crack is honest in one respect only: both are line-shaped records of a bulk transition, and both are found by looking where the grain went wrong.

Units keep the object honest. The tension is quoted as mass per length, an odd-looking unit that is simply energy per length with the speed of light squared divided out, and the numbers are stark: even the feeblest strings still allowed by the searches outweigh, per centimeter, a terrestrial mountain, while the heaviest ever proposed approach the mass of a star compressed into each hair's width. The reader who keeps the mass-per-length unit in mind will read every later limit correctly, because every fence in the subject is a fence on that one quantity.